SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

PROJECT NAME: REPLATE

GROUP MEMBERS: Aylin DÖNMEZ, Beyza Makbule YILDIRIM, Eren PAMUK, Hayrünisa TOPRAK, Hüseyin Alp ERCANDANOĞLU, Hüseyin ESEN, Ramazan YILDIRIM

TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Database Management System
2	User Interface / User Experience Design
3	Mapping and Location Services API

SOFTWARE TOOLS FOR TASK 1: Database Management System

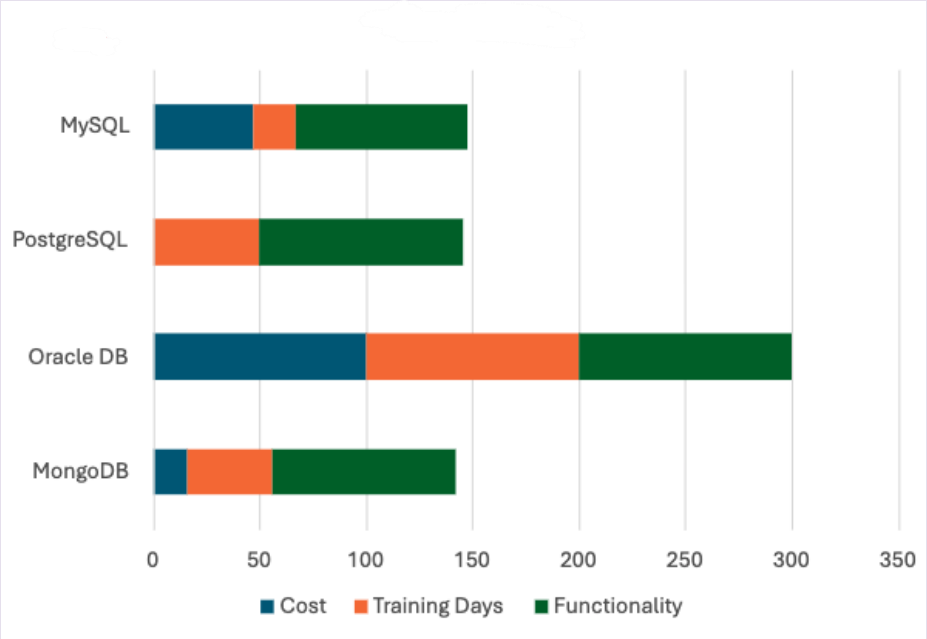
Tool Cost/Training/Functionality Data

Tool	MySQL	PostgreSQL	Oracle DB	MongoDB
Cost	$2250/y	$0	$4800/y	$784/y
Training Days	2	5	10	4
Functionality	80	95	99	85

Normalized Cost/Training/Functionality Data

Tool	MySQL	PostgreSQL	Oracle DB	MongoDB
Cost	46.9	0	**100**	16.3
Training Days	20	50	**100**	40
Functionality	80.8	96	**100**	85.9

Normalized Tool Graph



Which tool has been selected? Why?

For the RePlate project, we have selected PostgreSQL as our Database Management System (DBMS). As seen in the normalized tool graph, PostgreSQL provides the optimal balance by offering exceptionally high functionality (96) while keeping the software license cost completely free (0).

Although Oracle DB provides the highest raw functionality (100), it maximizes both the cost (100) and training days (100) indexes, making it highly inefficient and overly complex for our project's scope and budget. Conversely, MySQL and MongoDB have significantly lower functionality scores (80.8 and 85.9, respectively), which may not fully support our advanced system requirements. RePlate heavily relies on complex relational data—connecting users, businesses, reservations, and municipal distribution centers—as well as highly dependent location-based queries to display surplus food on an interactive map. PostgreSQL natively supports advanced spatial data processing through its PostGIS extension, making it the perfect fit to handle our map-based distance filtering and strict 25-meter GPS accuracy requirement. Furthermore, its moderate training requirement (50) ensures that our development team can adapt to it without causing severe delays, making it the most cost-effective and powerful solution for our system.

SOFTWARE TOOLS FOR TASK 2: User Interface / User Experience Design

Tool Cost/Training/Functionality Data

Tool	Framer	Figma	InVision	Sketch
Cost	$1200/y	$1000/y	$300/y	$500/y
Training Days	4	3	4	5
Functionality	95	100	80	85

Normalized Cost/Training/Functionality Data

Tool	Framer	Figma	InVision	Sketch
Cost	100	83.3	**25**	41.6
Training Days	80	60	**80**	100
Functionality	95	100	**80**	85

Normalized Tool Graph

Which tool has been selected? Why?

For the RePlate project, we selected Figma as our User Interface (UI) and User Experience (UX) Design tool. When analyzing the normalized data, Figma provides the absolute maximum functionality (100) while maintaining highly efficient training days (60) and a reasonable cost ratio (83.3) compared to premium tools like Framer.

Figma's cloud-based, real-time collaboration feature allows our entire team to work simultaneously on the same design files, regardless of whether they use Windows or macOS—a major limitation for tools like Sketch. Additionally, we have a strict non-functional requirement (NFR-6) stating that at least 90% of our users must be able to complete a product reservation in under 3 minutes. Figma's advanced interactive prototyping capabilities will allow us to simulate the entire reservation flow and conduct rapid usability testing with real users before writing any code. Therefore, Figma is the ultimate tool to guarantee an intuitive, fail-proof interface for our application

SOFTWARE TOOLS FOR TASK 3: Mapping and Location Services API

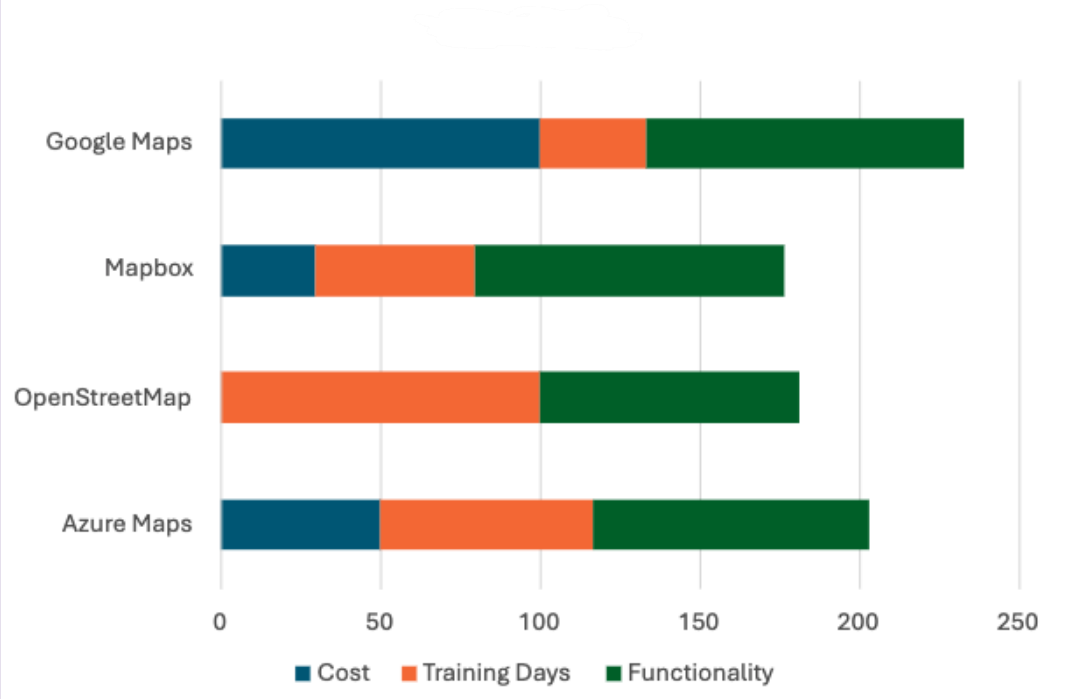
Tool Cost/Training/Functionality Data

Tool	Google Maps	Mapbox	OpenStreetM	Azure Maps
Cost	$2000/y	$600/y	$0	$1000/y
Training Days	2	3	6	4
Functionality	98	95	80	85

Normalized Cost/Training/Functionality Data

Tool	Google Maps	Mapbox	OpenStreetM	Azure Maps
Cost	100	30	**0**	50
Training Days	33.3	50	**100**	66.7
Functionality	100	96.9	**81.6**	86.7

Normalized Tool Graph



Which tool has been selected? Why?

For the RePlate project, we selected Mapbox as our Mapping and Location Services API. According to our normalized data, Mapbox provides an exceptional balance, offering near-perfect functionality (96.9) compared to Google Maps, but at a significantly lower cost (30 vs 100).

RePlate highly depends on interactive maps to display surplus food locations, filter products by distance, and guide users to municipal distribution centers. Mapbox provides highly customizable map interfaces that will seamlessly fit into our UI design. Furthermore, it easily meets our strict non-functional requirement (NFR-11) of maintaining a 25-meter maximum GPS error margin. While OpenStreetMap is free, its high training and setup time (100) makes it inefficient for our project schedule. Therefore, Mapbox is the most cost-effective, developer-friendly, and powerful mapping tool for RePlate.